



Terpenoids of *Ganoderma lucidum* reverse cognitive impairment through attenuating neurodegeneration via suppression of *PI3K/AKT/mTOR* expression in vivo model

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ABSTRACT

This study investigated effects of triterpenoids derived from *Ganoderma lucidum* (GLT) on the ability to attenuate cognitive impairment and reduce oxidative stress and inflammation induced by D-galactose using an aged rats model. The data revealed that dietary GLT intervention improved the memory of aged rats in a water maze experiment. A reduction in inflammation was also achieved as GLT decreased the levels of MDA, AGEs, NO, TNF- α and AChE activity, and increased the activity of T-AOC, GSH-Px, T-SOD, CAT and IL-2 level in serum. The GLT intervention revised hepatocyte cellular arrangement and binucleation of the liver status, and even hippocampal neurons morphology. GLT upregulated the expression of *FOXO4*, *SIRT1*, and downregulated the expression of *GFAP*, *iNOS*, *PI3K*, *AKT*, *mTOR* and *IL-6* with activation of *PI3K/AKT/mTOR* pathway. The current study highlights a promising dietary approach for prevention or alleviation of oxidative stress, inflammation and cognitive impairment induced by age-related conditions.

1. Introduction

Normal aging in human is linked to an increased incidence of diseases, including cancer, diabetes, cardiovascular and neurodegenerative disorders such as atherosclerosis and Alzheimers disease. *In vitro* and *in vivo* studies have shown that the cellular aging is a result of chronic oxidative damage caused by an increase in reactive oxygen species (ROS) and reduced levels of antioxidants (Naidoo & Birch-Machin, 2017). Ageing is associated with a multitude of cellular processes such as DNA damage, epigenetic alterations, loss of proteostasis, the deregulation of insulin and IGF-1 signaling, mitochondrial dysfunction, inflammation, cellular senescence which culminate in a reduced regenerative potential of tissues (López-Otín, Blasco, Partridge, Serrano, & Kroemer, 2013). The activation of NF- κ B signaling pathways can lead to aging and the inhibition of these pathways and has shown to extend lifespan (Kim & Choe, 2019). One animal model of drug induced and rapid ageing is the administration of D-galactose which results in increased oxidative stress as a result of the formation of advanced

glycation end products (AGEs) with associated deficits in learning and memory abilities (Chen et al., 2019). The lipid metabolism may also be dysregulated by D-galactose and may one of the mechanisms through which D-galactose can induce aging (Zhou et al., 2017). Recently, many studies have applied D-galactose to establish aging models to investigate the effects of drugs or certain active ingredients on improving aging.

Ganoderma lucidum (GL) is a medicinal fungus that is commonly used as traditional medicine to prevent and treat a variety of diseases for a long term in Asian countries. The GL has been proved to have potent immunomodulatory and anti-tumor effects (Lin & Zhang, 2004) and is associated with less side-effect compared with radiotherapy and chemotherapy treatment (Jin, Beguerie, Sze, & Chan, 2012). The GL contains a variety of active ingredients, of which triterpenoids and polysaccharides are the most significant and the most studied, while *Ganoderma lucidum* triterpenoids (GLT) directly inhibit the growth of cancer cells (Qu et al., 2017). Other research also reported that GLT can also suppress the production of tumor necrosis factor-alpha (TNF- α),

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interleukin-6 (IL-6) and nitric oxide (NO) from lipopolysaccharide-stimulated murine RAW264.7 cells (Dudhgaonkar, Thyagarajan, & Sliva, 2009). Anti-tumor and anti-metastatic effects in the liver of GLT have been shown to be associated with inhibition of tumor-induced angiogenesis (Kimura, Taniguchi, & Baba, 2002).

A range of bioactive compounds including carnosine, taurine, ursolic acid, salidroside, green tea polyphenols, ginsenoside Rg1, anthocyanins and quercetin have shown to improve the oxidative stress, inflammation, cell apoptosis, learning and memory abilities. Although there are some previous studies of GLT on the anti-tumors function as described above, the investigation of GLT intervention on the impact of hippocampal neurons is very limited. Thus, this study applied a D-galactose-induced aging rats model for investigating its effects on learning and memory abilities, and parameters of oxidative stress and inflammation status. More interestingly, hepatocyte cellular morphology, hippocampal neurons morphology and immunohistochemical analysis were also performed.

2. Materials and methods

2.1. Materials

GLT (Purity = 28%) was provided by Australian-Sino Biotechnology Co. Ltd (Longquan, Zhejiang, China). D-galactose and the vitamin C (Vc) were purchased from Beijing Solarbio Science & Technology Co., Ltd (Beijing, China).

2.2. Animal experiment

Sprague-Dawley (SD) rats (4-weeks old) were obtained from SPF (Beijing) Biotechnology Co., Ltd (Beijing, China). All rats were fed with a basic diet, and after one week's adaptive feed, the rats were randomly divided into five groups with 10 rats in each group. Control group (NC): subcutaneous injection of saline (0.2 mL) once a day; model group (MC): subcutaneous injection of D-galactose (300 mg/kg weight) (0.2 mL) once a day; positive control group (VC): subcutaneous injection of D-galactose (300 mg/kg weight) and Vc (20 mg/kg weight) by gavage once a day; low-dose intervention group (LGLT): subcutaneous injection of D-galactose (300 mg/kg weight) and GLT (10 mg/kg weight) by gavage once a day, and high-dose intervention group (HGLT): subcutaneous injection of D-galactose (300 mg/kg weight) and GLT (20 mg/kg weight) by gavage. These interventions were performed for four weeks and the rats were allowed to eat and drink freely. The surrounding environmental conditions were maintained at 23 °C with a 12 h light-dark cycle. After fasting for 12 h, blood and tissues were collected and stored at −80 °C.

2.3. Morris water maze trial

After four weeks of dietary intervention, rats were tested using a Morris water maze. The maze consisted of a tank (150 cm in diameter, 60 cm in height) filled with water at approximately 23 °C. The tank was divided into four quadrants, and an escape platform (12 cm in diameter, and 30 cm in height) was placed in the middle of the third quadrant. The height of water in the tank was maintained 1 cm higher than the escape platform. Each rat was placed in the pool at four different locations per day for four consecutive days. The rats were required to stand on the escape platform within 60 s, if not, the rats were placed on the escape platform and forced to memorize the surrounding environment. On the fifth day, probe test was performed by removing the escape platform and each rat was allowed to swim for 60 s. Latency to the escape platform, the number of crossing over the escape platform and the time in the different quadrants were measured. All data were recorded with a computerized video system (Morris water maze system, Huabei Zhenghua Biologic Instrument Equipment Co., Ltd, Huabei, China).

2.4. Analysis of serum parameters

Serum lipid composition and antioxidant capacity of total cholesterol (TC), triacylglycerol (TG), total superoxide dismutase (T-SOD), catalase (CAT), nitric oxide (NO), total antioxidant capacity (T-AOC) and glutathione peroxidase (GSH-Px) and the level of malonaldehyde (MDA) were measured using assay kits according to the manufacturers' instructions (Jiancheng Biological Engineering Institute, Nanjing, China). Neural activity, thinking and memory function of brain was measured using acetylcholine esterase (AChE) assay kit (Jiancheng Biological Engineering Institute, Nanjing, China). Measurement of age related degenerative diseases was assessed using a rat advanced glycation end products (AGEs) ELISA kit (Bio-Swamp Life Science Lab, Wuhan, China). The parameters associated with inflammation were measured using a rat tumor necrosis factor- α (TNF- α) ELISA kit and rat interleukin-2 (IL-2) ELISA kit according to the manufacturers' instructions (Zhuocai Biotechnology Co., Ltd, Shanghai, China).

2.5. Analysis of liver parameters

Liver tissue was homogenized with saline at the proportion of 1:9 (w/v) in an ice bath, and then centrifuged at 3000 r/m for 10 min. The supernatant was collected and the levels of MDA, GSH, GSH-PX, T-SOD and CAT were measured.

2.6. Histological analysis

Rat tissues were fixed in 10% formalin for 24 h at room temperature, and then washed using distilled water for 2–3 h following dehydration through a series of ethanol solutions, following treated with xylene to aid tissue transparency and then embedded in paraffin. Sections with 5 μ m were cut using a microtome and stained with hematoxylin and eosin (H&E), and the sections were analyzed by microscopy (Leica DM 1000 LED, Leica, Germany).

2.7. Immunohistochemistry analysis

The dried sections of 4 μ m were dewaxed with xylene, hydrated with a series of alcohols. Antigen was repaired by microwave heating, and then peroxide blocker (Beijing Biodragon Immunotechnologies Co., Ltd, Beijing, China) was added to remove the peroxide. After blocking with goat serum (Beijing Solarbio Science & Technology Co., Ltd, Beijing, China), a mouse anti-GFAP antibody (Santa Cruz Biotechnology Inc., Dallas, Texas, USA, 1:250 dilution) was added to the sections and incubated at 37 °C for 1 h. Goat anti-mouse antibody (Beijing Solarbio Science & Technology Co., Ltd, Beijing, China, 1:200 dilution) was added to the sections and incubated for 30 min at room temperature. After adding streptavidin (Beijing Solarbio Science & Technology Co., Ltd, Beijing, China) to remove biotin, the sections were developed by diaminobenzidine (DAB) developer (Beijing Solarbio Science & Technology Co., Ltd, Beijing, China) and counterstained by hematoxylin (Beijing Solarbio Science & Technology Co., Ltd, Beijing, China). After infiltration in a series of ethanol and xylene, the sections were sealed with neutral gum prior to the microscopy visualization.

2.8. RT-PCR analysis

Total RNA of hepatic and hippocampal tissues was extracted from the animal tissues (Tiangen Biochemical Technology Co., Ltd, Beijing, China), and reverse transcribed using a cDNA reverse transcription kit (Thermo Scientific Inc., Waltham, USA). The cDNA was stored at −80 °C. TB Green® Premix Ex Taq™ (Tli RNaseH Plus) (Takara, Beijing, China) and Real-Time PCR system (Applied Biosystems, Foster City, USA) were used for specific gene amplification using the following amplification conditions: 95 °C for 30 s followed by 40 cycles of 95 °C for 5 s and 60 °C for 30 s. Glyceraldehyde-3-phosphate dehydrogenase

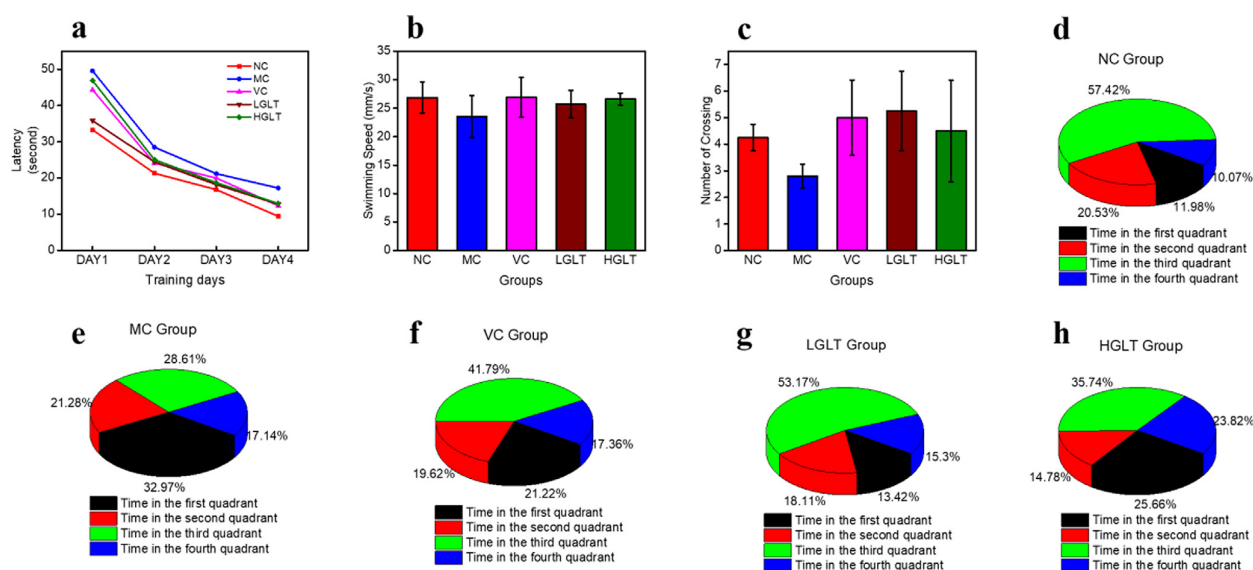


Fig. 1. Effect of GLT consumption on study and memory abilities in different animal groups. (a) Latency during training days. (b) Swimming speed on day 5. (c) The number of crossings on day 5. (d) Distribution of time consumed by rats in NC group in different quadrants. (e) Distribution of time consumed by rats in MC group in different quadrants. (f) Distribution of time consumed by rats in VC group in different quadrants. (g) Distribution of time consumed by rats in LGLT group in different quadrants. (h) Distribution of time consumed by rats in HGLT group in different quadrants.

(GAPDH) was used as an internal control. Primer sequence is presented in Table S1.

2.9. Statistical analysis

Results are shown as mean $\pm$ standard deviation (SD) values. Significant differences between groups were compared by one-way analysis of variance. When $p < 0.05$, the difference between the groups was considered statistically significant and $p < 0.01$ was considered extremely statistically significant. Values annotated with ^{abc}, no sharing a common letter indicates significant difference between groups ($p < 0.05$).

3. Results and discussion

3.1. Changes in the learning and memory behavior following GLT intervention

The ability to learn and lay down memory in aging rats was measured using the Morris water maze test. As shown in Fig. 1a, as the training time prolonged, the latency period in all groups gradually decreased. The latency of rats in the MC group was always higher than the rats in other groups, and the latency of rats in the NC group was always lowest among all the groups during the training days, highlighting the aging effect of D-Galactose. Through either Vc or GLT intervention, the latency decreased for the aged rats compared to the MC group, which appeared to restore the ability of D-Galactose treated rats to learn and produce memory. More interestingly, the improved effect was better in the LGLT group during the first training day, and similar results were obtained among the three intervention groups on day 4. Furthermore, data in Fig. 1b highlights the average swimming speed on day 5, in which the results showed a lower swimming speed in D-Galactose rat (MC group) compared with the NC group, and a faster swimming speed was obtained in the three intervention groups (VC, LGLT and HGLT groups). On day 5, the escape platform was removed and the areas within which the rats were activity were observed. The results showed that the number of crossings (Fig. 1c) was increased with a greater activity in the third quadrant (Fig. 1d-h) in the three intervention groups compared with MC group (D-Galactose only).

These results indicated that the intervention on the D-galactose-induced aging had a profound effect on the ability of rats to learn and memorize of the study, and in particular, the treatment with either Vc or GLT could restore these abilities.

3.2. Effect of GLT on bio-parameters of the serum samples

As animals age, lipid metabolism in tissues can become dysregulated and this can be evident in changes in serum biomarkers. Serum TC and TG levels tend to rise with age with a reduced metabolism in aging individuals (Yu et al., 2018). The results showed that the levels of TC (Fig. 2a) and TG (Fig. 2b) were increased through D-galactose treatment and then significantly reversed with feeding of either Vc or GLT. Lipids are important for a variety of biological processes, including aging and longevity (Papsdorf & Brunet, 2019). Decreasing TG levels through GLT treatment may contribute to and extending lifespan or improved health status during aging. This effect appeared to be dose dependent with HGLT showing a further enhanced beneficial effect on lipid metabolism.

The end product of lipid peroxidation, MDA, is considered as a biomarker of oxidative damage and previous studies have showed that aging is positively correlated with peroxidative damage (İnal, Kanbak, & Sunal, 2001). Following D-galactose treatment, the level of MDA increased and the activities of T-SOD, CAT and GSH-Px decreased compared to the control (Govindan et al., 2016). In this study, intervention from Vc, LGLT and HGLT significantly decreased the levels of MDA formation (Fig. 2c), while significantly increased the activity of the anti-oxidant level of T-SOD (Fig. 2f) and CAT (Fig. 2g). Furthermore, intervention from Vc, LGLT and HGLT also increased the level of T-AOC (Fig. 2d) and GSH-Px (Fig. 2e). Overall, the anti-oxidant response appeared to be regulated with a dose-dependent manner, in which rats in HGLT group demonstrated a repaired function on oxidative damage.

Meanwhile, AChE is a specific cholinergic marker, which is responsible for the degradation of acetylcholine in synaptic cleft, and used for reflecting the learning and memory capacity. In this study, the consumption of D-galactose significantly increased the activity of AChE (Fig. 2h), which was consistent with previous studies (Ji, Liu, Zhao, & Yu, 2015). In contrast, this study revealed that the intervention from Vc

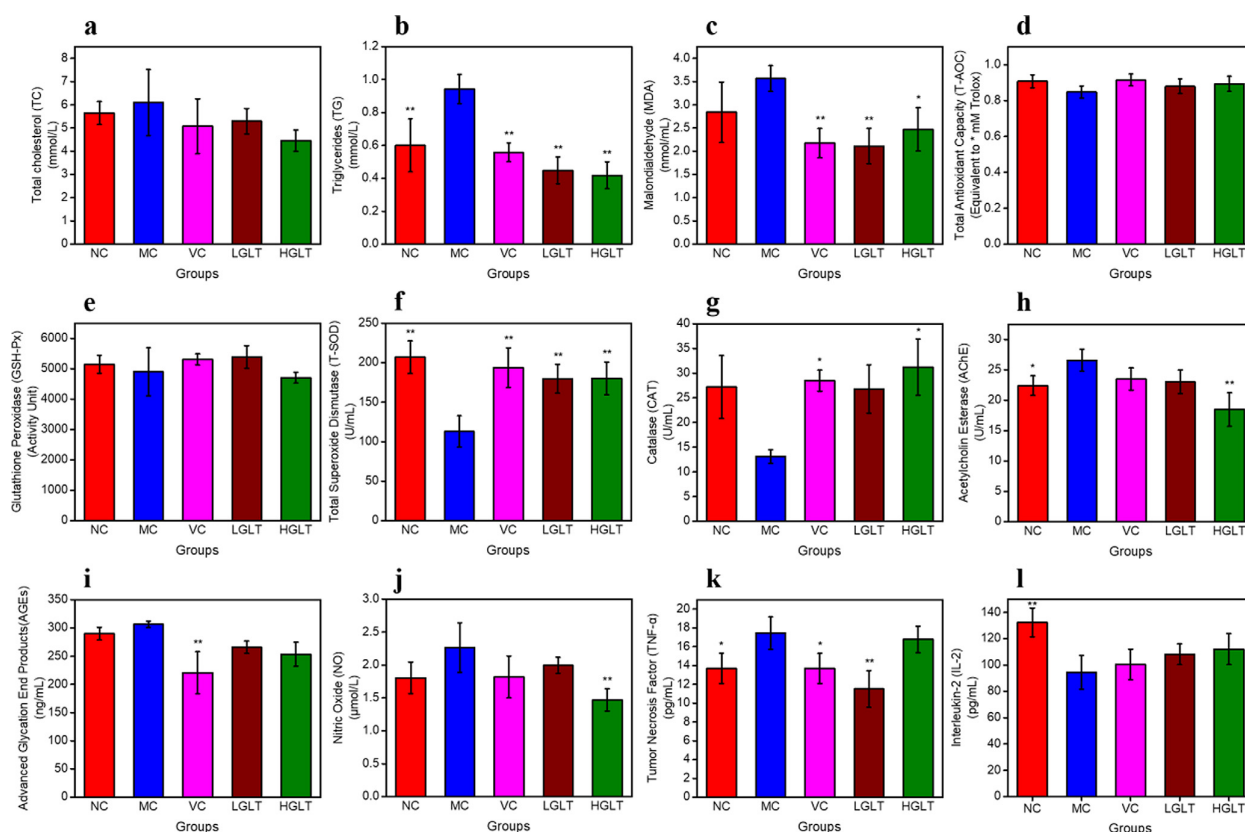


Fig. 2. Effect of GLT on lipids composition and anti-oxidant capacity of serum samples in different animal groups. (a) TC; (b) TG; (c) MDA; (d) T-AOC; (e) GSH-Px; (f) T-SOD; (g) CAT; (h) AChE; (i) AGEs; (j) NO; (k) TNF-α; (l) IL-2. TC: total cholesterol; TG: triacylglycerol; MDA: malondialdehyde; T-AOC: total antioxidant capacity; GSH-Px: glutathione peroxidase; T-SOD: total superoxide dismutase; CAT: catalase; AChE: acetylcholine esterase; AGEs: advanced glycation end products; NO: nitric oxide; TNF-α: tumor necrosis factor-alpha; IL-2: interleukin-2. ** indicates $p < 0.05$ and *** indicates $p < 0.01$, respectively.

and GLT reduced the AChE activity, and this reduction was also significant in the HGLT group compared to the MC group.

The proteins of AGEs are related to aging and the level of AGEs has been shown to accumulate in tissues due to chronic diseases associated with aging, such as atherosclerosis, diabetes, arthritis and neurodegenerative diseases (Ott et al., 2014). Thus, the inhibition of AGEs formation could limit the oxidative and inflammatory damage to tissues and subsequently contribute to delaying the development of pathophysiology, and improving the quality of life during aging (Baynes, 2001). In this study, the intake of Vc significantly decreased the level of AGEs, while the intervention from HGLT achieved a greater reduction in AGEs in serum when compared to LGLT (Fig. 2i).

NO is a marker of central nervous system inflammation, and high concentrations of NO are neurotoxic that induce defect in spatial memory and diminish the ability of the cells to deal with oxidative stress (Duan et al., 2017). The intervention of both Vc and GLT reduced the content of NO, and the reduction was significant in the HGLT group compared to the MC group (Fig. 2j).

Aging is considered to be associated with chronic inflammation, in which TNF-α is an inflammatory cytokine produced and its presence in serum is related to tissue damage. Furthermore, IL-2 is an immune signaling molecule that plays a role in T cell regulation, and helps maintain self-tolerance and prevent autoimmune diseases. Alteration in IL-2 function can potentially lead to autoimmune diseases (Trotta et al., 2018). Previous studies have shown that serum TNF-α increases (Puchta et al., 2016) and IL-2 reduces (Kim et al., 2018) following the aging process. The current study noted that the intake of either Vc or GLT significantly reduced serum TNF-α level (Fig. 2k) compared to the MC group, in particular in LGLT group. However, rats in HGLT group

achieved the best results on the restore of the level of serum IL-2 (Fig. 2l) among the intervention groups.

3.3. Different intervention on liver oxidative stress

It has been documented that liver is a major organ involved in D-galactose metabolism and as a result this treatment can cause oxidative stress in the liver (Feng et al., 2016). As shown in Table 1, compared with NC group, the hepatic MDA content increased by 44.83% and hepatic GSH content, GSH-Px, T-SOD and CAT activities decreased by 18.29%, 1.95%, 3.09%, 10.47%, respectively. In contrast, GLT reduced these oxidative damages induced by D-galactose. In HGLT group, the data showed that the content of MDA was decreased by 9.52%, GSH increased by 22.39% and the activities of GSH-Px, T-SOD and CAT increased by 34.81%, 16.59% and 27.33%, respectively compared to MC group. Thus, the result suggested that the dietary intake of triterpenoids improved the T-AOC and potentially contribute to a reversal of a mild fatty liver condition (Chiu et al., 2017).

3.4. Changes in histopathological property in liver and hippocampus tissues following intervention

Data in Fig. 3 shows the different histopathological property of the liver and hippocampus in the different groups. Hepatocytes in the NC group were well organized, whereas the hepatocytes in the MC group became disordered, in which binucleation and vacuoles appeared on the liver of the MC group. Through the intervention of vitamin C and GLT, the binucleation of hepatocytes reduced, and a more order cellular appearance was achieved.

Table 1
Effect of GLT on the content of MDA and antioxidant activity in the liver of aging rats.

	NC	MC	VC	LGLT	HGLT
MDA (nmol/mgprot)	0.58 ± 0.09 ^a	0.84 ± 0.10 ^c	0.74 ± 0.06 ^b	0.80 ± 0.04 ^{bc}	0.76 ± 0.04 ^{bc}
GSH (mgGSH/gprot)	0.82 ± 0.08 ^b	0.67 ± 0.08 ^a	0.79 ± 0.08 ^b	0.76 ± 0.04 ^a	0.82 ± 0.07 ^b
GSH-Px (Activity Unit)	1038.7 ± 115.1 ^a	1018.4 ± 45.1 ^a	1490.6 ± 45.4 ^d	1241.2 ± 108.1 ^b	1372.9 ± 86.4 ^c
T-SOD (U/mgprot)	923.6 ± 29.3 ^a	895.0 ± 19.6 ^a	1143.6 ± 27.6 ^{bc}	1209.9 ± 133.0 ^c	1043.5 ± 116.7 ^{ab}
CAT (U/mgprot)	41.9 ± 6.3 ^{ab}	37.5 ± 4.1 ^a	52.3 ± 6.8 ^c	46.3 ± 2.6 ^{bc}	47.7 ± 2.8 ^{bc}

Results were expressed as means ± SD (one-way ANOVA). Different superscript lowercase letters in the same row indicate significant difference ($p < 0.05$).

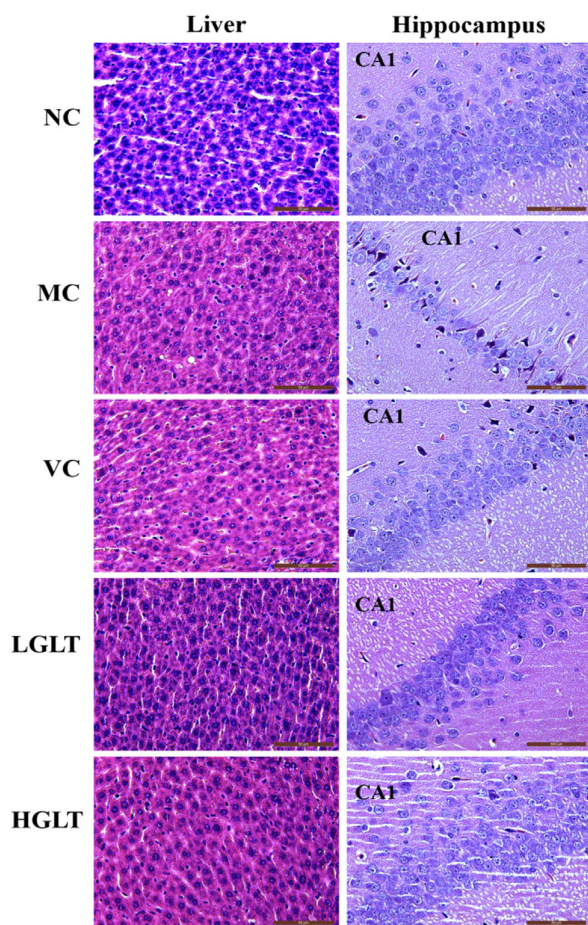


Fig. 3. Effect of GLT on the histopathological changes in liver and hippocampus.

The role of the hippocampus is mainly responsible for storing information and is a key part of learning and memory, and it consists of four main histological subregions, CA1, CA2, CA3, and the dentate gyrus (DG). The region of CA1 plays an important role in the retrieval of memory and of spatial memory (Stackman, Cohen, Lora, & Rios, 2016). The hippocampal neurons in NC group were abundant and each of them were full in the CA1 area. In contrast, in the MC group, it seems that the number of hippocampal neurons was decreased and the hippocampal neurons showed acidophily degeneration from the visualization (Fig. 3), and these abnormalities of hippocampal neurons were consistent with previous research (Li, Gong, Wu, Lu, & Shi, 2010). Through the intervention of Vc and GLT, these abnormalities were reversed. In

particular, the HGLT intervention returned the hippocampal region to a normal state in terms of both the number and morphology of hippocampal neurons (Fig. 3).

3.5. Analysis of GFAP in hippocampus

Glial fibrillary acidic protein (GFAP) is a marker of astrocyte activation, mainly distributed in astrocytes of the central nervous system. In aging and inflammatory organisms, increased astrocytic reactivity can be observed (Palmer & Ousman, 2018). Fig. 4 showed the expression of GFAP in hippocampus of different groups. The expression of GFAP was increased in MC group while decreased through Vc and GLT intervention. And Vc and GLT can revise the expression of GFAP induced by D-galactose significantly, which showed the effect of anti-aging and anti-inflammation.

3.6. Analysis of liver and hippocampus gene expression

The forkhead box (FOX) transcription factors that belong to the other (O) subfamily (FOXO) include four members (FOXO1, 3, 4, 6) in mammals. The FOXO transcription factors regulate cellular differentiation, growth, survival, cell cycle, metabolism, stress and tumor suppression pathways (Tia et al., 2018). The current study revealed that the expression of FOXO4 gene in liver was decreased by D-galactose treatment and significantly increased through Vc and GLT intervention, especially HGLT (Fig. 5a). Previous research also showed that FOXO4-null mice were more susceptible to trinitrobenzene sulfonic acid (TNBS) injury-induced colitis and FOXO4 could inhibit nuclear factor- κ B (NF- κ B) (Zhou et al., 2009).

Furthermore, the expression of phosphatidylinositol 3-kinase (PI3K), protein kinase B (AKT) and mammalian target of rapamycin (mTOR) was found to be increased with the treatment of D-galactose. In contrast, these genes were decreased in different degrees following the intervention of either Vc or GLT (Fig. 5a, b). Literature review indicated that D-galactose led to aging in tissues, including liver and hippocampus (Chen, Chen, & Zhou, 2018), while GLT revised the state of tissues by reducing apoptosis (Wang, Cao, Zhao, & Feng, 2017). Inhibition of PI3K, AKT and mTOR could prevent or suppress cellular aging. Among them, PI3K promotes the phosphorylation and activation of AKT, and PI3K/AKT pathway mediates the effect of various neurogenic growth factors (Matsuda et al., 2018). PI3K and AKT were also found to be altered by aging (Matsuda et al., 2018). The mTOR pathway plays a role in the regulation of disease and aging in animal models through regulating cell growth and division (Kim & Choe, 2019).

Interventions from both Vc and GLT intervention promoted the expression of SIRT1 in liver (Fig. 5a). Sirtuins consist of a family of NAD⁺-dependent deacylases that are central to the body's response to diet and exercise. SIRT1 activation shows promise for the treatment of aging and age-related diseases. The overexpression of SIRT1 in brain could extend the lifespan of mice and the overexpression of SIR2 has been shown to extend the lifespan of yeast, *Caenorhabditis elegans* and

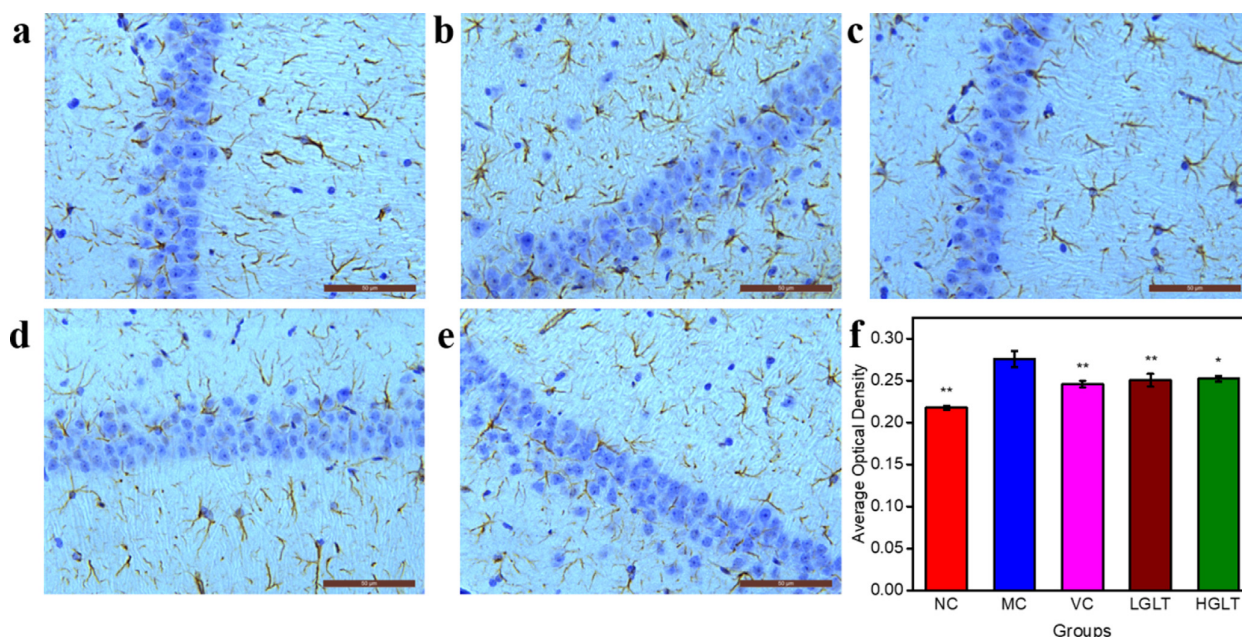


Fig. 4. Effect of GLT on the expression of GFAP in hippocampus. (a) The expression of GFAP in NC group. (b) The expression of GFAP in MC group. (c) The expression of GFAP in VC group. (d) The expression of GFAP in LGLT group. (e) The expression of GFAP in HGLT group. (f) The quantitative analysis of GFAP in the five groups. “*” indicates $p < 0.05$ and “***” indicates $p < 0.01$, respectively.

drosophila (Park, Yoo, & Park, 2017). *SIRT1* could regulate numerous vital signaling pathways, including DNA repair and apoptosis, neurogenesis and cell stress responses (Elibol & Kilic, 2018).

More interestingly, in this study, the intervention from GLT significantly reduced the expression of inducible nitric oxide synthase (*iNOS*) in hippocampus (Fig. 5b). The *iNOS* is regulated by multiple transcription factors, and upregulated in the state of inflammation and oxidative stress. Consistently, the consumption of D-galactose upregulated the expression of *iNOS* which was paralleling with previous research (Shwe, Pratchayasakul, Chattipakorn, & Chattipakorn, 2018). Aging is accompanied by the increased expression of *iNOS*, and the lack of *iNOS* could not prevent aging-associated insulin resistance (Cha et al., 2010).

The current brain study also indicated that the expression of *IL-6* was enhanced in the aged brain and GLT could reduce the expression of *IL-6* significantly in the hippocampus (Fig. 5b). It has been reported that the overexpression of *IL-6* in brain could inhibit memory and learning

capacity, causing neurodegeneration and exacerbate pathology induced by other cytokines (Kinney et al., 2018).

Overall, this study found that GLT could revise pathology associated with aging in the liver and hippocampus and this mode of action may be via inhibiting the expression of *mTOR*, *PI3K*, *AKT*, *iNOS* and *IL-6* and also upregulating the expression of *FOXO4* and *SIRT1*.

4. Conclusion

Aging is associated with oxidative stress, inflammation and cell apoptosis. Through the Morris water maze test, this study indicated that GLT could increase the number of crossings and time spent in the third quadrant. The intervention of GLT also showed its functionality on the improving of learning and memory and inflammation capacity. Through the measurement of serum lipids composition and oxidative stress/anti-oxidant capacity, GLT also demonstrated the ability to attenuate the lipid metabolism disorder induced by the D-galactose, and

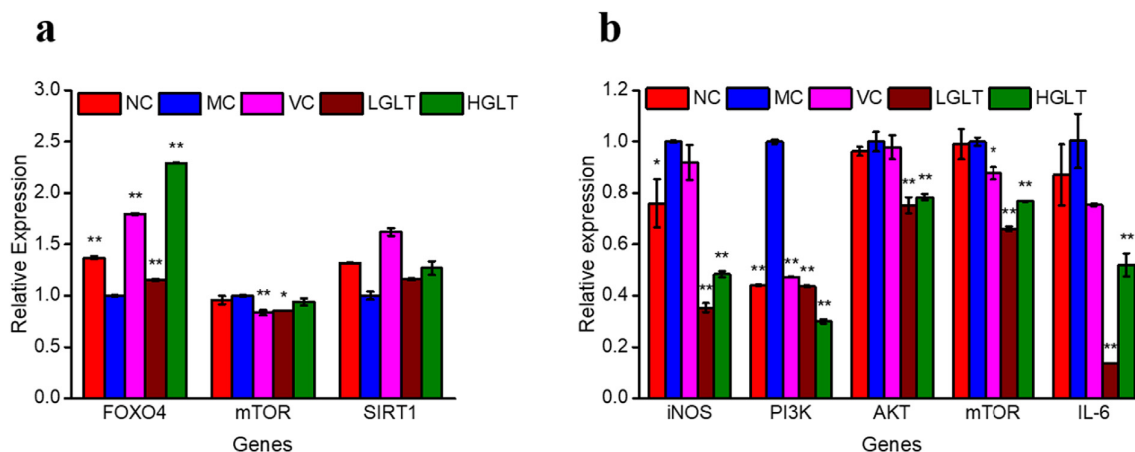


Fig. 5. Gene expression of liver and hippocampus. (a) liver; (b) hippocampus. “*” indicates $p < 0.05$ and “***” indicates $p < 0.01$, respectively.

this effect is characterized with a dose-dependence. It was further proved via the analysis of liver histopathology. More interestingly, GLT intervention displayed similar roles on the improvement of the aging related syndromes in hippocampal tissues, which may imply their association with each other of the organs. This might be the first report to reveal that the presence of GLT in the diet improved the expression of *FOXO4*, *mTOR*, *SIRT1* in the liver and GFAP, *iNOS*, *PI3K*, *AKT*, *IL-6* in the hippocampus.

5. Ethics statement

All animal experiments were approved by the Ethical Committee for the Experimental Use of Animals at the Center for Drug Safety Evaluation, Tianjin University of Science and Technology (Approval No. A3375/707).

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CRediT authorship contribution statement

Anqi Wang: Methodology, Investigation, Writing - original draft. **Changhao Xiao:** Software, Data curation. **Jianxian Zheng:** Conceptualization, Methodology, Resources. **Chuansong Ye:** Methodology, Resources. **Zhen Dai:** Software, Validation. **Qinghai Wu:** Software, Formal analysis. **Jinguang Liu:** Investigation. **Padraig Strappe:** Writing - review & editing. **Zhongkai Zhou:** Writing - review & editing, Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jff.2020.104142>.

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